

20TH ANNUAL API CYBERSECURITY CONFERENCE FOR THE NATURAL GAS AND OIL INDUSTRY

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Mental Health In Cybersecurity:
Utilizing the Copenhagen Burnout Inventory
Cybersecurity Version



American
Petroleum
Institute



THE PRESSURE COOKER

“You wouldn’t want my job any week”
- Tommy Norris

The Pressure Cooker

- We aren't printing t-shirts over here. This is hard. There's no "clock out" time.
- We're understaffed and under-resourced. Then, RIF'd.
- We're not paranoid. Everyone is out to get us.
- We never disconnect, because we can't face the outcome if there is a breach while we're gone.
- "Digital Transformation" is killing us. Relabeling the same tech over and over isn't helping, either.

Digital Transformation Impacts

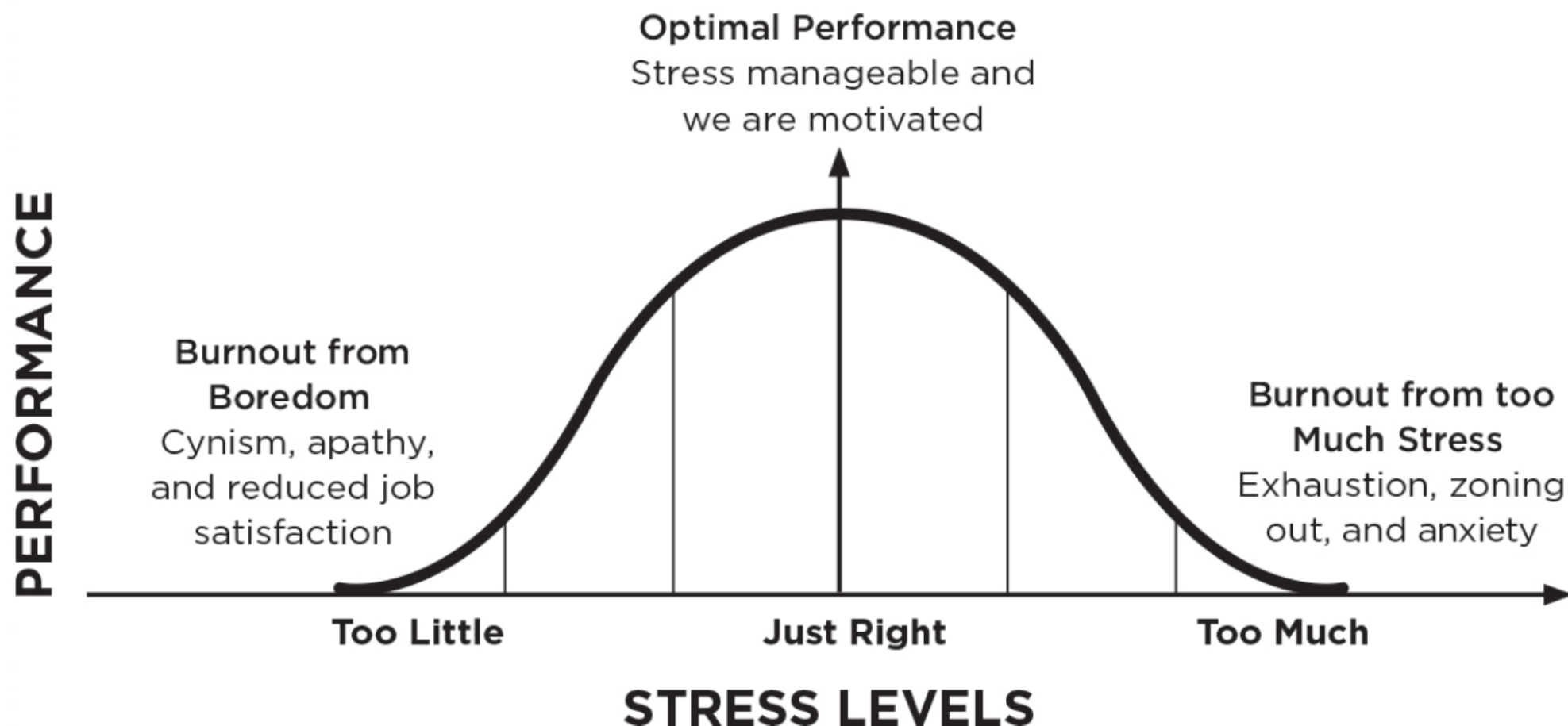
The research backs that up:

87.5% of DT projects fail (Wade & Shan, 2020). That failure rate doesn't just mean lost budgets—it means teams left holding the bag after high-stakes, high-visibility efforts collapse, creating burnout loops.

DT is being sold as a performance lift, but each transformation type (mobility, analytics, integrated systems, etc.) comes with behavioral, operational, and technical challenges—most of which land directly on the teams expected to sustain them.

Is All Stress Bad? No.

Yerkes-Dodson Law





01

Recognizing Burnout and Imposter Syndrome

Is This You?

Many professionals carry the weight of burnout, believing they alone are responsible for causing it.

Worse, they believe they alone must fix it.

Wrong.

Burnout isn't a reflection of your weakness. It's not something you can overcome with sheer willpower or working harder.

Think about it: We didn't build stronger canaries to prevent deaths in coal mines. We built safer mines.

Burnout is a shared responsibility, and real change starts with creating conditions that allow people to thrive.

Frenetic Burnout

This describes the pattern of working harder and harder in response to stress. It happens when the demands of work outweigh the resources needed to do it comfortably and there's an ambition to do it well.

Your work–life boundaries therefore slip, and work becomes all encompassing.

Those who care passionately about what they do are at higher risk of this, particularly where budget cuts and shrinking margins have led to a lack of staff, squeezing those who are left

"The Trauma of Burnout"

Frenetic Burnout

- Research calls out that **behavioral challenges** (user adaptation, cultural shift) make up **46% of all failure points** in DT projects—that's *people problems*, not tech problems.
- “Digital Mindset” sounds inspiring, but in practice it means **continuous reskilling at scale** while keeping the lights on. For overburdened security teams, that's a formula for frenetic burnout.
- Compounded DT (multiple intertwined projects) impacts teams exponentially.

Frenetic Burnout

Emerging Technologies in Digital Transformation:

Intelligence & Automation

Artificial Intelligence (AI)

Machine Learning

Predictive Analytics

Core Infrastructure & Platforms

Cloud Computing

Big Data

Analytics

Examples

24/7 Operations: Pipeline leaks don't wait for business hours.

High-Stakes Downtime: Shutting down a refinery due to cyber risk carries national economic consequences.

Dual Accountability: Operators balance **worker safety and cybersecurity** simultaneously.

Amplified Pressure Cooker: Burnout is magnified by risks to **lives, environment, and national energy supply**.

The Evolution

Initial Drive: CISOs, IT, and Cybersecurity professionals often start with a strong drive to succeed, fueled by a desire to prove their capabilities, and achieve recognition.

Imposter Syndrome: As they advance, many experience imposter syndrome, doubting their abilities despite evidence of their competence due to the rapid change in the industry. As they continue to improve, the industry evolution exceeds their own.

Continual Drive: The need to constantly prove themselves, due to the additional roles being placed on them, they push hard, which pushes professionals past their limits.

Burnout: Prolonged stress can lead to burnout, validating the negative beliefs and insecurities they've carried, and potentially leading to a cycle of self-doubt and further burnout.

Burnout Is More Common Than You Think

Tom Hanks

"No matter what we've done, there comes a point where you think, 'How did I get here? When are they going to discover that I am, in fact, a fraud and take everything away from me?'"

Patrick Kelley

I own two successful cybersecurity companies. Each day, I start with a healthy dose of imposter syndrome.

My ruminations continually convince me that I'll never be deserving enough to be here.

Cognitive Distortion, unchecked, will steal your happiness.

Are You Facing Burnout?

- I often worry that people will discover I'm not as capable as they think I am.
- I frequently attribute my successes to luck or external factors rather than my abilities.
- I find it difficult to accept praise or compliments about my work.
- I often feel I have to work harder than others to prove my worth.
- When I make mistakes, I see it as proof of my inadequacy rather than a normal learning process.



02

BUTTERFLY
EFFECT

(CHAOS THEORY)

The Blast Radius Of Burnout

Studies shows that advanced DT projects (those combining 2–3 types) have higher upside but also massively higher risk of collapse.

When these fail or stall, there's a hidden human cost:

- Institutional knowledge is burned out of the org.
- Staff are left in a “permanent incident response” posture - exactly what we see in security teams post-breach.
- Turnover spikes, increasing the operational fragility of the team.

Examples

Colonial Pipeline Breach: A case study in how a single incident created cascading national consequences.

Government Mandates: Post-incident directives increased compliance burdens, adding to operator stress.

Permanent Incident Response: Many teams remain in a constant “post-breach posture,” never fully decompressing.

Burnout as a Risk Multiplier: Fatigue-driven mistakes in OT environments can lead to **physical safety incidents** or **critical supply disruptions**.

Self-Neglect – Where It Ends

“I can’t step away, the team needs me.”

This most often occurs post-event. A teammate will often stay in the fight, because they won't leave other team members "behind".

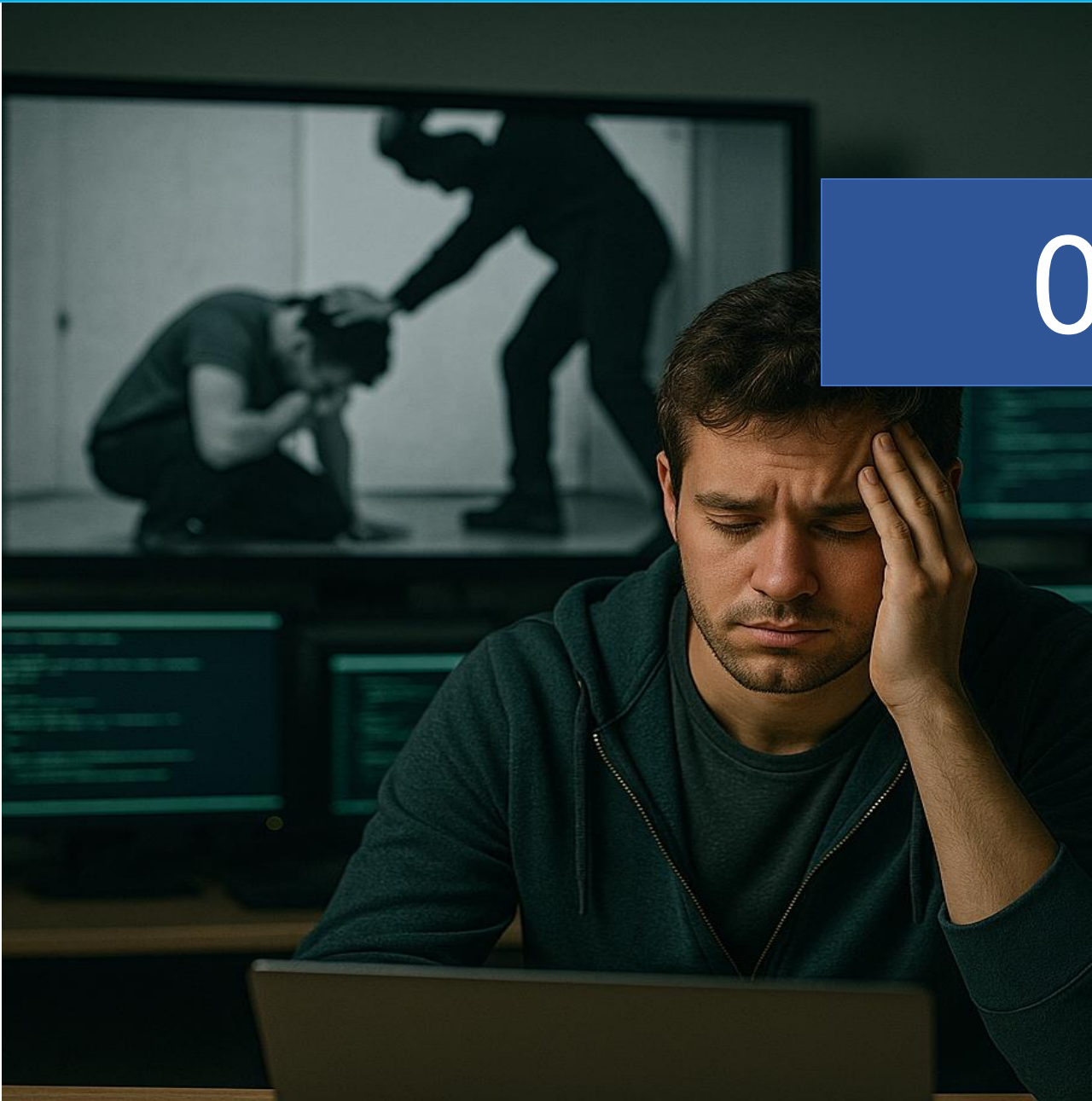
“I’m the only one who understands this environment.”

Post-breach, the teammate becomes hypervigilant, especially if a breach has ever occurred on their watch.

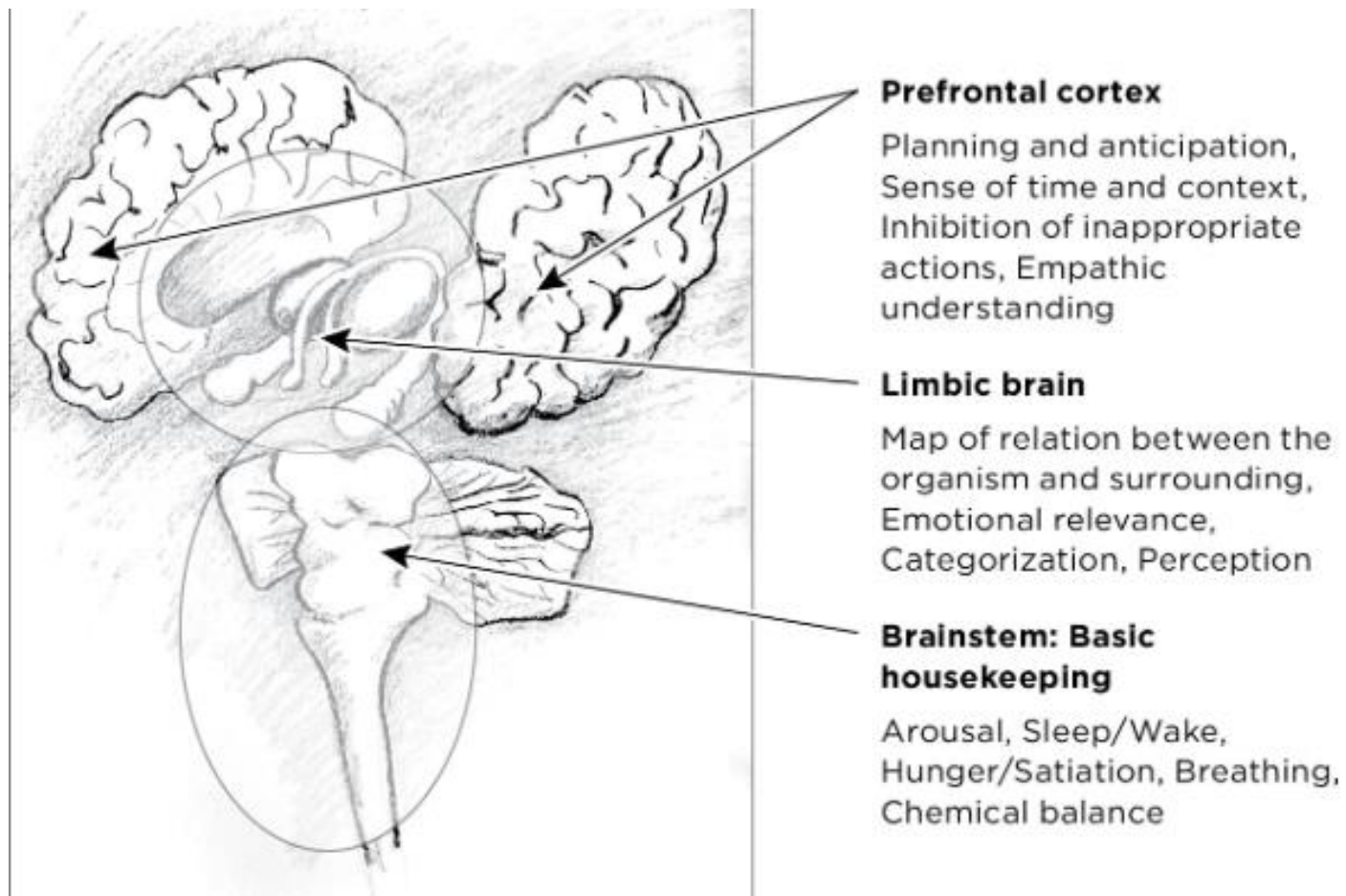
The learn the environment so well that they can "hear footsteps" and are wary to pass it to another team member.

03

The Trauma- Burnout Connection



Shared Physiology And Neurobiology



Cybersecurity – Constant Fight or Flight

- Images and sounds are stored in our brain, especially with trauma. They aren't always stored in order.
- fMRI scan results show that triggering a memory that pulls those images and sounds actually erases all of the time since the event.
- Therapy takes extended time, because the rational brain is nearly impotent in talking the emotional brain out of its own reality.

Strengths & Vulnerabilities

Skill Developed Through Trauma	Benefit	Burnout Risk
Hypervigilance	Strong Threat Detection	Constant Anxiety, Difficulty Relaxing
Emotional Regulation	Maintains Composure In Stress	Suppresses Emotions, Leading To Exhaustion
Pattern Recognition	Identifying Anomalies	Mental Fatigue From Overanalyzing Threats
Resilience & Adaptability	Thrives Under Pressure	Pushing Too Hard, Neglecting Self-Care

Longterm Impacts

Cortisol and HPA Axis Dysregulation

Chronic stress leads to **persistent elevation or dysregulation of cortisol**. Initially, cortisol is **elevated (hypercortisolemia)** to manage stress.

Over time, the system can become **blunted (hypocortisolemia)** due to negative feedback exhaustion.

Impacts:

- Impaired immune function (increased inflammation).
- Disrupted sleep-wake cycles.
- Memory and learning deficits.

Self-Reflection

When was the last time you were fully relaxed
when you looked at your phone messages?

When you went on your last vacation or break,
how long did it actually take you to "unwind"?



04

Coping Mechanisms



We Aren't Doing A Great Job

The stress most cyber professionals are under includes work hours, role ambiguity, role conflict, role overload, and more.

Nearly every Slack team associated with Cyber has a channel called "Drink" or "The Bar". Few have one for "Mental Health".

2 in 5 CISOs report abusing or excessively drinking multiple nights per week to offset stress.

Why Do We Lean On Alcohol?

A key effect of alcohol is that it enhances gamma-aminobutyric acid (GABA), a neurotransmitter responsible for inhibiting our responses.

Or to put it another way, GABA slows down our thoughts and reactions. When thoughts are racing at 100 miles an hour, this calming effect is exactly what we crave, so it's no wonder that we reach for the quickest (and often socially accepted) way of achieving it.

With time, we develop a tolerance for the sedative effects of alcohol and start to require more of it to achieve the same calming results.

This leads to insomnia as the body struggles to process the amount of sugar in the consumed alcohol.

Coping Mechanisms

Industry Realities: Shift work, remote postings, and long rotations create unique mental health pressures.

Cultural Normalization: Post-shift heavy drinking is often accepted as a coping mechanism.

Hidden Cost: Reliance on alcohol can worsen fatigue, mental health decline, and safety risks.

Healthier Alternatives:

Structured decompression practices

Employee Assistance Programs (EAP)

Peer support networks designed for field and plant teams

Who's Working For Who?

It's important to regularly ask yourself,

"Are these coping methods still serving me, or am I serving them?"

If it's the latter, it's time to reach out and talk to someone.

Self- Reflection

How can you cope better?



05

Understanding Burnout Measurement

Why Measure?

Quantification leads to measured and meaningful change:

- Measurement translates feelings of stress into clear, trackable metrics.
- Tracking burnout over time allows early identification of rising trends before they become critical incidents.
- Measuring enables leaders to take targeted, data-driven actions to improve team health.



06

Copenhagen Burnout Inventory

Consider The Copenhagen Burnout Inventory

Why CBI?

- Designed for cyber operators and defenders.
- Measures operational stress within SOC/IR context.
- Non-punitive, team-focused, and actionable.

How is it measured?

- Emotional Exhaustion (feeling drained, tired post-incident)
- Cynicism & Detachment (users as problems, emotional distance)
- Personal Accomplishment (meaningful work, success in defense)

What's Included

Burnout Calculators & Guides

- Burnout Calculator – Run Guide.docx
- Burnout Cost Calculator – Run Guide.docx
- CBI-CS_Burnout_Calculator.xlsx
- CBI-CS_Cost_Calculator.xlsx

CBI-CS Materials

- CBI-CS_Implementation and Administration Guide.docx
- CBI-CS_Questionnaire.docx
- CBI-CS_Training_Guide.docx

What's Included

Communication & Feedback

- Communication Plan for Burnout Assessment Findings.docx
- Follow-up and Feedback Mechanism.docx

Support & Intervention

- Employee Support Resources.docx
- Intervention and Mitigation Strategy Document.docx

Policies

- Privacy and Ethical Handling Policy.docx

Download The Program

<https://cbi-cs.com>

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QUESTIONS?
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